CSME SCGM

Analytical & Numerical Solutions to the Non-Diffusive Stefan Problem

Matthew Van Ham[a] · Samuel Huberman[a,b] · Minghan Xu[c]

[a] Dept. of Chemical Engineering, McGill University
[b] Dept. of Physics, McGill University
[c] Dept. of Civil & Mineral Engineering, University of Toronto

McGill

Background (1/2): The Stefan Problem

Solid-liquid phase change with a moving boundary: a classical free-boundary PDE.

Applications:

- Permafrost thaw
- Microscale melting
- Droplet freezing
- Phase change materials (PCMs)

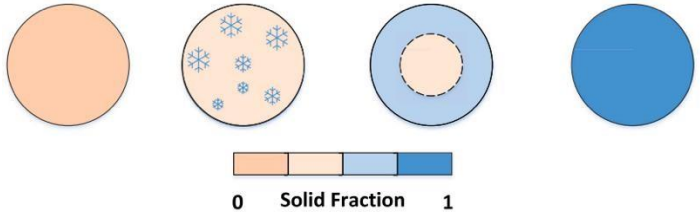


Figure 1: Application of the Stefan problem (droplet freezing).

The Stefan number:

$$\mathrm{Ste} = \frac{c_p(T_\mathrm{f} - T_0)}{L^*}$$

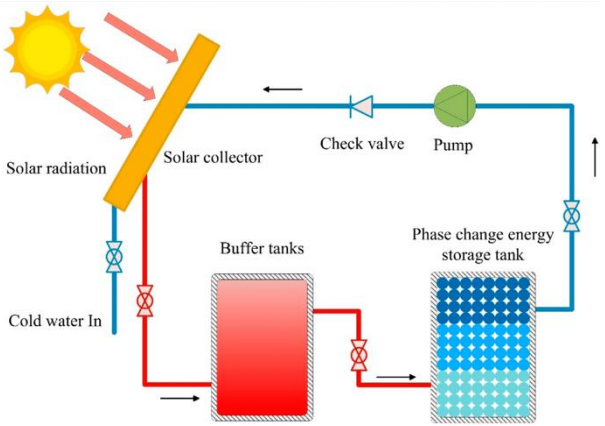


Figure 2: Application of the Stefan problem (phase change materials).

What we're looking at:

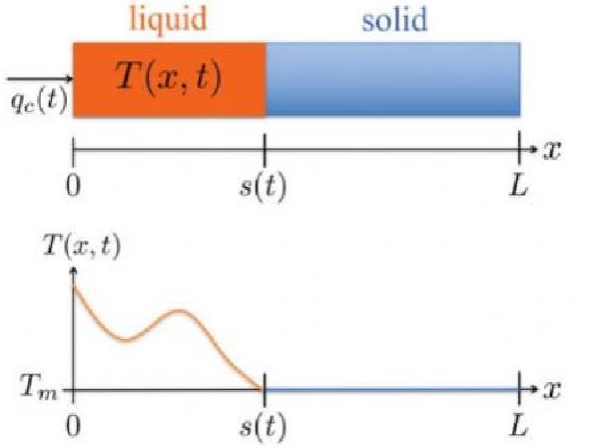


Figure 3: Phase diagram and temperature profile of general 1D, rectangular domain Stefan Problem.

Classical solution (one-phase, semi-infinite, diffusive):

$$T(x,t) = T_0 + (T_\mathrm{f} - T_0)\frac{\mathrm{erf}(\frac{x}{2\sqrt{\alpha t}})}{\mathrm{erf}(\lambda)}$$

$$x_\mathrm{i}(t) = 2\lambda\sqrt{\alpha t}$$

What if we consider non-diffusive effects?

Background (2/2): Non-Fourier Heat Transfer

Fourier Heat Transfer

$$q(x,t) = -k\frac{\partial T}{\partial x}$$

Characteristics:

- Diffusive behavior
- Infinite information propagation
- Macroscopic manifestation of statistical dynamics
- $\tau = 0$, no relaxation period

Non-Fourier Heat Transfer

$$q(x,t+\tau) = -k\frac{\partial T}{\partial x}$$

Characteristics:

- Non-Diffusive behavior
- Finite thermal wave speed ($c = \sqrt{\frac{\alpha}{\tau}}$)
- $\tau \neq 0$, thermal relaxation time delay

Classical Fourier's law breaks down at nanoscales or ultrafast heating
(When: $\mathcal{O}(t) \approx \mathcal{O}(\tau)$)

$$q(x,t+\tau) = q(x,t) + \tau\frac{\partial q}{\partial t} + \mathcal{O}(\tau^2)$$

$$\frac{\partial^2 T}{\partial x^2} = \frac{1}{\alpha}\frac{\partial T}{\partial t}, \quad \Rightarrow \quad \frac{\partial^2 T}{\partial x^2} = \frac{\tau}{\alpha}\frac{\partial^2 T}{\partial t^2} + \frac{1}{\alpha}\frac{\partial T}{\partial t}, \qquad 0 < x < x_i(t), \quad t > 0,$$

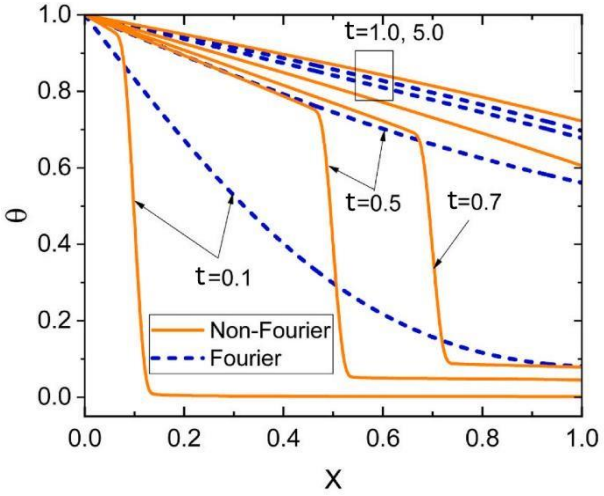


Figure 4: Dimensionless temperature profile vs dimensionless position for Fourier and non-Fourier heat transfer models (Kundu et al., *Thermal Engineering,* 2025).

Key Idea: Perturbation Expansion in the Stefan Number

Stefan number controls the problem's deviation from the simple Ste = 0 case. When Ste $\ll 1$, the solution is a small perturbation (Locally linear) from a simple base state, enabling a systematic asymptotic expansion.

$$\frac{\partial^2 T}{\partial x^2} = \frac{\tau}{\alpha}\frac{\partial^2 T}{\partial t^2} + \frac{1}{\alpha}\frac{\partial T}{\partial t}, \qquad 0 < x < x_{\mathrm{i}}(t), \quad t > 0, \qquad \Rightarrow \qquad \frac{\partial^2 \widetilde{T}}{\partial \widetilde{x}^2} = \widetilde{\tau}\,\mathrm{Ste}\,\frac{\partial^2 \widetilde{T}}{\partial \widetilde{x}_{\mathrm{i}}^2}\left(\frac{\mathrm{d}\widetilde{t}}{\mathrm{d}\widetilde{x}_{\mathrm{i}}}\right)^{-2} + \mathrm{Ste}\,\frac{\partial \widetilde{T}}{\partial \widetilde{x}_{\mathrm{i}}}\left(\frac{\mathrm{d}\widetilde{t}}{\mathrm{d}\widetilde{x}_{\mathrm{i}}}\right)^{-1}, \qquad 0 < \widetilde{x} < \widetilde{x}_{\mathrm{i}},$$

1

Non-Dimensionalization

$$\widetilde{\tau} = \frac{\alpha\tau}{X^2} \times \mathrm{Ste}, \quad \widetilde{T} = \frac{T - T_0}{T_{\mathrm{f}} - T_0}, \quad \widetilde{x} = \frac{x}{X}, \quad \widetilde{x}_{\mathrm{i}} = \frac{x_{\mathrm{i}}}{X},$$

$$\mathrm{Ste} = \frac{c_p(T_{\mathrm{f}} - T_0)}{L^*}, \quad \widetilde{t} = \frac{\alpha t}{X^2} \times \mathrm{Ste},$$

2

Expand in Powers of Ste

$$\widetilde{T} \sim \widetilde{T}_0 + \mathrm{Ste}\,\widetilde{T}_1 + \mathrm{Ste}^2\,\widetilde{T}_2 + \cdots,$$

$$\widetilde{t} \sim \widetilde{t}_0 + \mathrm{Ste}\,\widetilde{t}_1 + \mathrm{Ste}^2\,\widetilde{t}_2 + \cdots.$$

3

Solve Order by Order

Leading order (Ste→0) gives a linear ODE; first-order correction adds hyperbolic effects.

Analytic Formulation: Perturbation Expansions

Prescribed Flux BC (Fourier):

$$\left.\frac{\partial \widetilde{T}}{\partial \widetilde{x}}\right|_{\widetilde{x}=\widetilde{x}_\mathrm{i}} = \left(\frac{\mathrm{d}\widetilde{t}}{\mathrm{d}\widetilde{x}_\mathrm{i}}\right)^{-1},$$

$$\frac{\partial^2 \widetilde{T}}{\partial \widetilde{x}^2} = \widetilde{\tau}\,\mathrm{Ste}\,\frac{\partial^2 \widetilde{T}}{\partial \widetilde{x}_\mathrm{i}^2}\left(\frac{\mathrm{d}\widetilde{t}}{\mathrm{d}\widetilde{x}_\mathrm{i}}\right)^{-2} + \mathrm{Ste}\,\frac{\partial \widetilde{T}}{\partial \widetilde{x}_\mathrm{i}}\left(\frac{\mathrm{d}\widetilde{t}}{\mathrm{d}\widetilde{x}_\mathrm{i}}\right)^{-1}, \qquad 0 < \widetilde{x} < \widetilde{x}_\mathrm{i},$$

Leading Order:

$$\frac{\partial^2 \widetilde{T}_0}{\partial \widetilde{x}^2} = 0, \qquad 0 < \widetilde{x} < \widetilde{x}_\mathrm{i},$$

First Order:

$$\frac{\partial^2 \widetilde{T}_1}{\partial \widetilde{x}^2} = \widetilde{\tau}\,\frac{\partial^2 \widetilde{T}_0}{\partial \widetilde{x}_\mathrm{i}^2}\left(\frac{\mathrm{d}\widetilde{t}_0}{\mathrm{d}\widetilde{x}_\mathrm{i}}\right)^{-2} + \frac{\partial \widetilde{T}_0}{\partial \widetilde{x}_\mathrm{i}}\left(\frac{\mathrm{d}\widetilde{t}_0}{\mathrm{d}\widetilde{x}_\mathrm{i}}\right)^{-1}, \qquad 0 < \widetilde{x} < \widetilde{x}_\mathrm{i},$$

Series Solution:

$$\widetilde{T}(\widetilde{x},\widetilde{t}) \sim -1 + \frac{\widetilde{x}}{\widetilde{x}_\mathrm{i}} + \mathrm{Ste} \times \left[\frac{\widetilde{\tau}\widetilde{x}^3}{3\widetilde{x}_\mathrm{i}^5} + \frac{-2\widetilde{\tau}\widetilde{x} - \widetilde{x}^3}{6\widetilde{x}_\mathrm{i}^3} + \frac{\widetilde{x}}{6\widetilde{x}_\mathrm{i}}\right] + \ldots$$

$$\widetilde{t}(\widetilde{x}_\mathrm{i}) \sim \frac{\widetilde{x}_\mathrm{i}^2}{2} + \mathrm{Ste} \times \left[\frac{2\widetilde{x}_\mathrm{i}^2}{3} - \frac{2}{3}\widetilde{\tau}\ln\widetilde{x}_\mathrm{i} + c\right] + \ldots$$

Analytic Formulation: Perturbation Expansions

Prescribed Flux BC (Fourier):

$$\left.\frac{\partial \widetilde{T}}{\partial \widetilde{x}}\right|_{\widetilde{x}=\widetilde{x}_\mathrm{i}} = \left(\frac{\mathrm{d}\widetilde{t}}{\mathrm{d}\widetilde{x}_\mathrm{i}}\right)^{-1},$$

$$\frac{\partial^2 \widetilde{T}}{\partial \widetilde{x}^2} = \widetilde{\tau}\,\mathrm{Ste}\,\frac{\partial^2 \widetilde{T}}{\partial \widetilde{x}_\mathrm{i}^2}\left(\frac{\mathrm{d}\widetilde{t}}{\mathrm{d}\widetilde{x}_\mathrm{i}}\right)^{-2} + \mathrm{Ste}\,\frac{\partial \widetilde{T}}{\partial \widetilde{x}_\mathrm{i}}\left(\frac{\mathrm{d}\widetilde{t}}{\mathrm{d}\widetilde{x}_\mathrm{i}}\right)^{-1}, \qquad 0 < \widetilde{x} < \widetilde{x}_\mathrm{i},$$

Leading Order:

$$\frac{\partial^2 \widetilde{T}_0}{\partial \widetilde{x}^2} = 0, \qquad 0 < \widetilde{x} < \widetilde{x}_\mathrm{i},$$

First Order:

$$\frac{\partial^2 \widetilde{T}_1}{\partial \widetilde{x}^2} = \widetilde{\tau}\,\frac{\partial^2 \widetilde{T}_0}{\partial \widetilde{x}_\mathrm{i}^2}\left(\frac{\mathrm{d}\widetilde{t}_0}{\mathrm{d}\widetilde{x}_\mathrm{i}}\right)^{-2} + \frac{\partial \widetilde{T}_0}{\partial \widetilde{x}_\mathrm{i}}\left(\frac{\mathrm{d}\widetilde{t}_0}{\mathrm{d}\widetilde{x}_\mathrm{i}}\right)^{-1}, \qquad 0 < \widetilde{x} < \widetilde{x}_\mathrm{i},$$

Series Solution:

$$\widetilde{T}(\widetilde{x},\widetilde{t}) \sim -1 + \frac{\widetilde{x}}{\widetilde{x}_\mathrm{i}} + \mathrm{Ste} \times \left[\frac{\widetilde{\tau}\widetilde{x}^3}{3\widetilde{x}_\mathrm{i}^5} + \frac{-2\widetilde{\tau}\widetilde{x} - \widetilde{x}^3}{6\widetilde{x}_\mathrm{i}^3} + \frac{\widetilde{x}}{6\widetilde{x}_\mathrm{i}}\right] + \ldots$$

$$\widetilde{t}(\widetilde{x}_\mathrm{i}) \sim \frac{\widetilde{x}_\mathrm{i}^2}{2} + \mathrm{Ste} \times \left[\frac{2\widetilde{x}_\mathrm{i}^2}{3} - \frac{2}{3}\widetilde{\tau}\ln\widetilde{x}_\mathrm{i} + c\right] + \ldots$$

Analytic Formulation: Rectification of Asymptotic Divergence

Outer Solution (Fourier BC):

$$\left.\frac{\partial \widetilde{T}}{\partial \widetilde{x}}\right|_{\widetilde{x}=\widetilde{x}_\mathrm{i}} = \left(\frac{\mathrm{d}\widetilde{t}}{\mathrm{d}\widetilde{x}_\mathrm{i}}\right)^{-1},$$

$$\frac{\partial^2 \widetilde{T}}{\partial \widetilde{x}^2} = \widetilde{\tau}\,\mathrm{Ste}\,\frac{\partial^2 \widetilde{T}}{\partial \widetilde{x}_\mathrm{i}^2}\left(\frac{\mathrm{d}\widetilde{t}}{\mathrm{d}\widetilde{x}_\mathrm{i}}\right)^{-2} + \mathrm{Ste}\,\frac{\partial \widetilde{T}}{\partial \widetilde{x}_\mathrm{i}}\left(\frac{\mathrm{d}\widetilde{t}}{\mathrm{d}\widetilde{x}_\mathrm{i}}\right)^{-1}, \qquad 0 < \widetilde{x} < \widetilde{x}_\mathrm{i},$$

Leading Order:

$$\frac{\partial^2 \widetilde{T}_0}{\partial \widetilde{x}^2} = 0, \qquad 0 < \widetilde{x} < \widetilde{x}_\mathrm{i},$$

First Order:

$$\frac{\partial^2 \widetilde{T}_1}{\partial \widetilde{x}^2} = \widetilde{\tau}\,\frac{\partial^2 \widetilde{T}_0}{\partial \widetilde{x}_\mathrm{i}^2}\left(\frac{\mathrm{d}\widetilde{t}_0}{\mathrm{d}\widetilde{x}_\mathrm{i}}\right)^{-2} + \frac{\partial \widetilde{T}_0}{\partial \widetilde{x}_\mathrm{i}}\left(\frac{\mathrm{d}\widetilde{t}_0}{\mathrm{d}\widetilde{x}_\mathrm{i}}\right)^{-1}, \qquad 0 < \widetilde{x} < \widetilde{x}_\mathrm{i},$$

Series Solution:

$$\widetilde{T}(\widetilde{x},\tilde{t}) \sim -1 + \frac{\widetilde{x}}{\widetilde{x}_\mathrm{i}} + \mathrm{Ste} \times \left[\frac{\widetilde{\tau}\widetilde{x}^3}{3\widetilde{x}_\mathrm{i}^5} + \frac{-2\widetilde{\tau}\widetilde{x} - \widetilde{x}^3}{6\widetilde{x}_\mathrm{i}^3} + \frac{\widetilde{x}}{6\widetilde{x}_\mathrm{i}}\right] + \ldots$$

$$\tilde{t}(\widetilde{x}_\mathrm{i}) \sim \frac{\widetilde{x}_\mathrm{i}^2}{2} + \mathrm{Ste} \times \left[\frac{2\widetilde{x}_\mathrm{i}^2}{3} - \frac{2}{3}\widetilde{\tau}\ln\widetilde{x}_\mathrm{i} + c\right] + \ldots$$

Fix #1: Inner Solution (Fourier BC)

$$\widetilde{t} = \sqrt{\mathrm{Ste}}\,\phi,$$

$$\frac{\partial^2 \widetilde{T}}{\partial \widetilde{x}^2} = \widetilde{\tau}\,\frac{\partial^2 \widetilde{T}}{\partial \phi^2} + \sqrt{\mathrm{Ste}}\,\frac{\partial \widetilde{T}}{\partial \phi}, \qquad 0 < \widetilde{x} < \widetilde{x}_\mathrm{i}(\phi),$$

Leading order:

$$\widetilde{T}_{0,\mathrm{in}}(\widetilde{x},\phi) = -1 + \frac{\widetilde{x}}{\widetilde{x}_\mathrm{i}} + \sum_{n=1}^{\infty}\left[\frac{2}{\pi n}\cos\left(\frac{n\pi c\,\phi}{\widetilde{x}_\mathrm{i}}\right) + B_n \sin\left(\frac{n\pi c\,\phi}{\widetilde{x}_\mathrm{i}}\right)\right]\sin\left(\frac{n\pi\widetilde{x}}{\widetilde{x}_\mathrm{i}}\right)$$

$$\frac{\mathrm{d}\phi_0}{\mathrm{d}\widetilde{x}_\mathrm{i}} = \left[\sqrt{\mathrm{Ste}}\,\frac{1}{\widetilde{x}_\mathrm{i}} + \frac{1}{\widetilde{x}_\mathrm{i}}\sum_{n=1}^{\infty}(-1)^n\left[2\cos\left(\frac{n\pi c\,\phi}{\widetilde{x}_\mathrm{i}}\right) + n\pi B_n \sin\left(\frac{n\pi c\,\phi}{\widetilde{x}_\mathrm{i}}\right)\right]\right]^{-1}$$

First Order:

$$\widetilde{T}_{s1}(\widetilde{x},\phi) = \widetilde{T}_{s0}(\widetilde{x},\phi) + \frac{1}{2c}\int_0^{\phi}\int_{\widetilde{x}-c(\phi-\eta)}^{\widetilde{x}+c(\phi-\eta)}\left[\widetilde{T}_{s0}(\psi,\eta)e^{\eta/(2\widetilde{\tau})} + 1 - \frac{\psi}{\widetilde{x}_i}\right]\mathrm{d}\psi\,\mathrm{d}\eta.$$

$$\left.\frac{d\phi(\widetilde{x},\phi)}{d\widetilde{x}_i}\right|_{\widetilde{x}=\widetilde{x}_i} = \left(\frac{\partial}{\partial\widetilde{x}}\left[\tilde{T}_{s0}(\tilde{x},\phi) + \frac{\partial}{\partial\tilde{x}}\frac{1}{2c}\int_0^{\phi}\int_{\widetilde{x}-c(\phi-\eta)}^{\widetilde{x}+c(\phi-\eta)}\left[\tilde{T}_{s0}(\psi,\eta)exp\left(\frac{\eta}{2\tilde{\tau}}\right) + \left(1 - \frac{\psi}{x_i}\right)\right]\right)^{-1}$$

Analytic Formulation: Rectification of Asymptotic Divergence

... Continutation of Inner Solution

A Composite Solution May be Constructed:

$$t_{\text{comp}}(\tilde{x}_i) = u(\tilde{x}_i)\, t_{\text{inner}}(\tilde{x}_i) + \left[1 - u(\tilde{x}_i)\right] t_{\text{outer}}(\tilde{x}_i),$$

$$0 \le u(\tilde{x}_i) \le 1, \qquad \lim_{\tilde{x}_i \to 0} u(\tilde{x}_i) = 1, \qquad \lim_{\tilde{x}_i \to 1} u(\tilde{x}_i) = 0$$

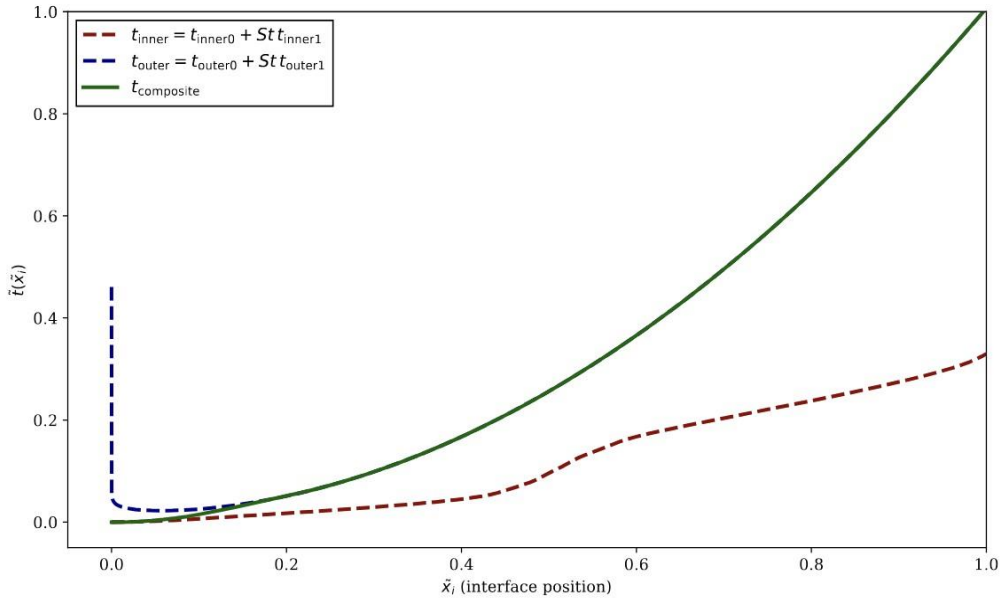


Figure 5: Composite Solution of Inner and Outer Interface Profile with 5th Order Polynomial blending Function $u(\tilde{x}_i^{2s})$ $(\alpha = 1.0, Ste = 0.001, \text{Pl} = 1.0)$

Analytic Formulation: Rectification of Asymptotic Divergence

... Continuation of Inner Solution

A Composite Solution May be Constructed:

$$t_{\mathrm{comp}}(\tilde{x}_i) = u(\tilde{x}_i)\, t_{\mathrm{inner}}(\tilde{x}_i) + \left[1 - u(\tilde{x}_i)\right] t_{\mathrm{outer}}(\tilde{x}_i),$$

$$0 \le u(\tilde{x}_i) \le 1, \qquad \lim_{\tilde{x}_i \to 0} u(\tilde{x}_i) = 1, \qquad \lim_{\tilde{x}_i \to 1} u(\tilde{x}_i) = 0$$

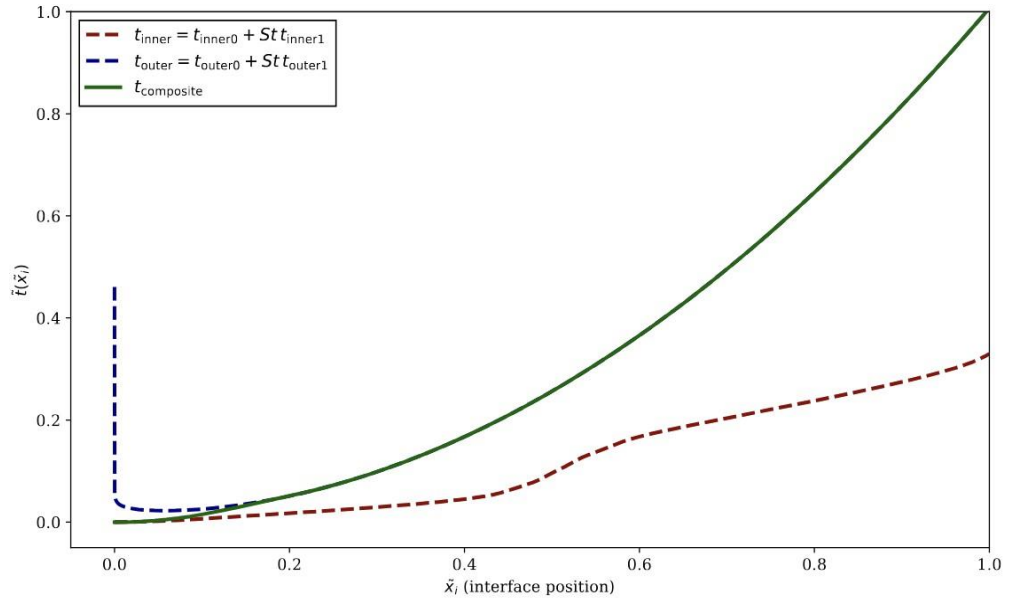


Figure 5: Composite Solution of Inner and Outer Interface Profile with 5th Order Polynomial blending Function $u(\tilde{x}_i^{2s})$ $(\alpha = 1.0, Ste = 0.001, \tilde{t} = 1.0)$

Fix #2: non-Diffusive Stefan Condition (non-Fourier BC)

$$L(t) = L^* \times \left[1 - \frac{p_1}{x_{\mathrm{i}}(t)}\right]^{p_2}, \qquad T_{\mathrm{f}}(t) = T_{\mathrm{f}}^* \times \left[1 - \frac{p_3}{x_{\mathrm{i}}(t)}\right]^{p_4}$$

When considering depressive model for Latent heat:

$$\left.\frac{\partial \tilde{T}}{\partial \tilde{x}}\right|_{\tilde{x}=\tilde{x}_{\mathrm{i}}} = \left(\frac{\mathrm{d}\tilde{t}}{\mathrm{d}\tilde{x}_{\mathrm{i}}}\right)^{-1} + \left[1 - \frac{1}{X\tilde{x}_{\mathrm{i}}}\right]\frac{1}{L^*}\left(\frac{\mathrm{d}\tilde{t}}{\mathrm{d}\tilde{x}_{\mathrm{i}}}\right)^{-1},$$

Leading Order:

$$\frac{\partial^2 \tilde{T}_0}{\partial \tilde{x}^2} = 0, \qquad 0 < \tilde{x} < \tilde{x}_{\mathrm{i}},$$

First Order:

$$\frac{\partial^2 \tilde{T}_1}{\partial \tilde{x}^2} = \tilde{\tau}\,\frac{\partial^2 \tilde{T}_0}{\partial \tilde{x}_{\mathrm{i}}^2}\left(\frac{\mathrm{d}\tilde{t}_0}{\mathrm{d}\tilde{x}_{\mathrm{i}}}\right)^{-2} + \frac{\partial \tilde{T}_0}{\partial \tilde{x}_{\mathrm{i}}}\left(\frac{\mathrm{d}\tilde{t}_0}{\mathrm{d}\tilde{x}_{\mathrm{i}}}\right)^{-1}, \qquad 0 < \tilde{x} < \tilde{x}_{\mathrm{i}},$$

Series Solution:

$$\tilde{T}(\tilde{x}, \tilde{t}) \sim -1 + \frac{\tilde{x}}{\tilde{x}_{\mathrm{i}}} + \mathrm{Ste} \times \left[\frac{L^* X(\tilde{x}\tilde{x}_{\mathrm{i}}^2 - \tilde{x}^3)[\tilde{x}_{\mathrm{i}}(X\tilde{x}_{\mathrm{i}} - 1) + L^* X(\tilde{x}_{\mathrm{i}}^2 - 2\tilde{\tau})]}{6\tilde{x}_{\mathrm{i}}^3[(1 + L^*)X\tilde{x}_{\mathrm{i}} - 1]^2}\right] + \ldots$$

$$\tilde{t}(\tilde{x}_{\mathrm{i}}) \sim \frac{(1 + L^*)\tilde{x}_{\mathrm{i}}^2}{2L^*} - \frac{\tilde{x}_{\mathrm{i}}}{XL^*} + \mathrm{Ste} \times \left[\frac{2\tilde{x}_{\mathrm{i}}^2}{3} - \frac{2L^*\tilde{\tau}\ln[1 + (1 + L^*)X\tilde{x}_{\mathrm{i}}]}{3(1 + L^*)}\right] + \ldots$$

Numerical verification: Verifying Analytical Solutions

Scheme:
TVD MacCormack with numerical diffusion
(Second order spatial and temporal accuracy)

$$\frac{\partial}{\partial x}\left(\mu(x)\frac{\partial U}{\partial x}\right) \approx D_{i+1/2} - D_{i-1/2}$$

Features:
- Diffusion helps capture the wave-fronts (Figure 3 (a), (b))
- Convergent over the Courant number and mesh size (Figure 4 (a), (b))

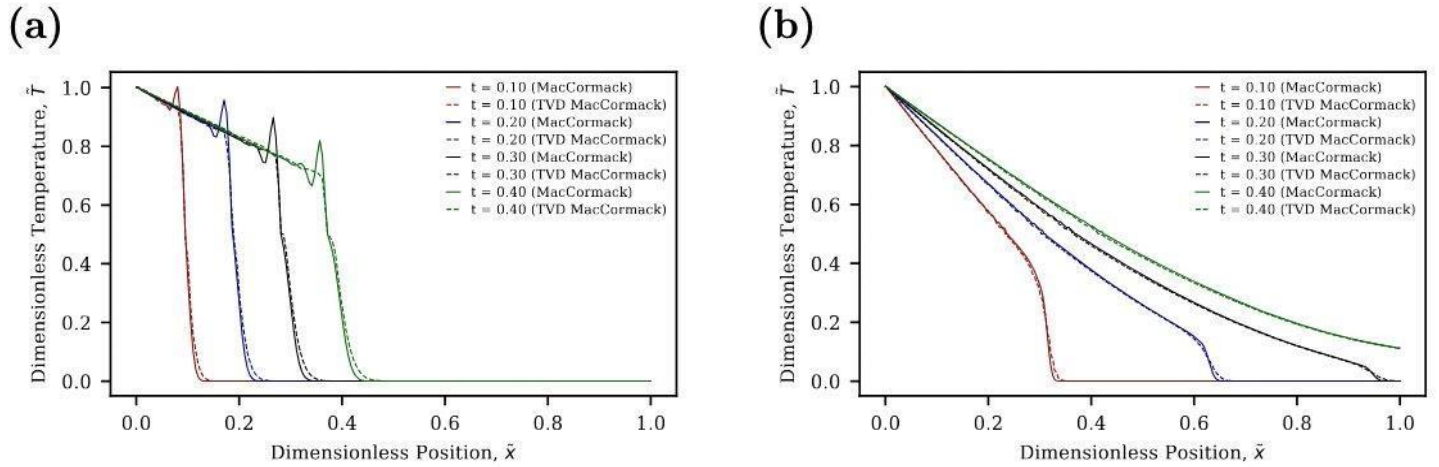


Figure 6: Dimensionless Temperature vs. Dimensionless Position for MacCormack-TVD and Standard MacCormack Methods for (a) High Thermal Relaxation Time ($\alpha = 1.0, Ste = 10.0, \tilde{n} = 1.0$) and (b) Low Thermal Relaxation Time ($\alpha = 1.0, Ste = 10.0, \tilde{n} = 0.1$)

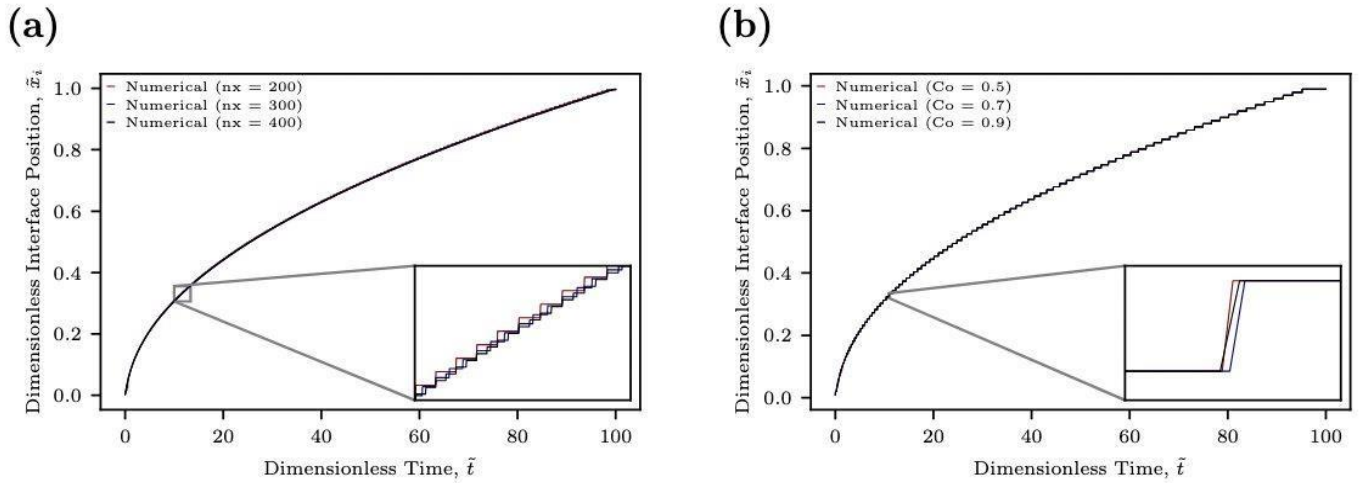


Figure 7: Dimensionless Interface Position vs. Dimensionless Time for (a) different mesh sizes ($\alpha = 1.0, Ste = 0.01, \tilde{n} = 1.0$) and (b) different Courant numbers ($\alpha = 1.0, Ste = 0.01, \tilde{n} = 1.0$)

Results and Parametric Studies

Ste

- Error across profile scales with Ste^2 (truncation error of series expansion)
- Inner composite solution error scales with Ste

τ

- Better early-time alignment with non-Fourier Stefan Condition
- Small effect on error, local linear approximation in temperature is exact in parabolic solution

α

- Temporal transformation (consider t dimensionalization) enhances early-time relative error
- No major contribution to later-time error

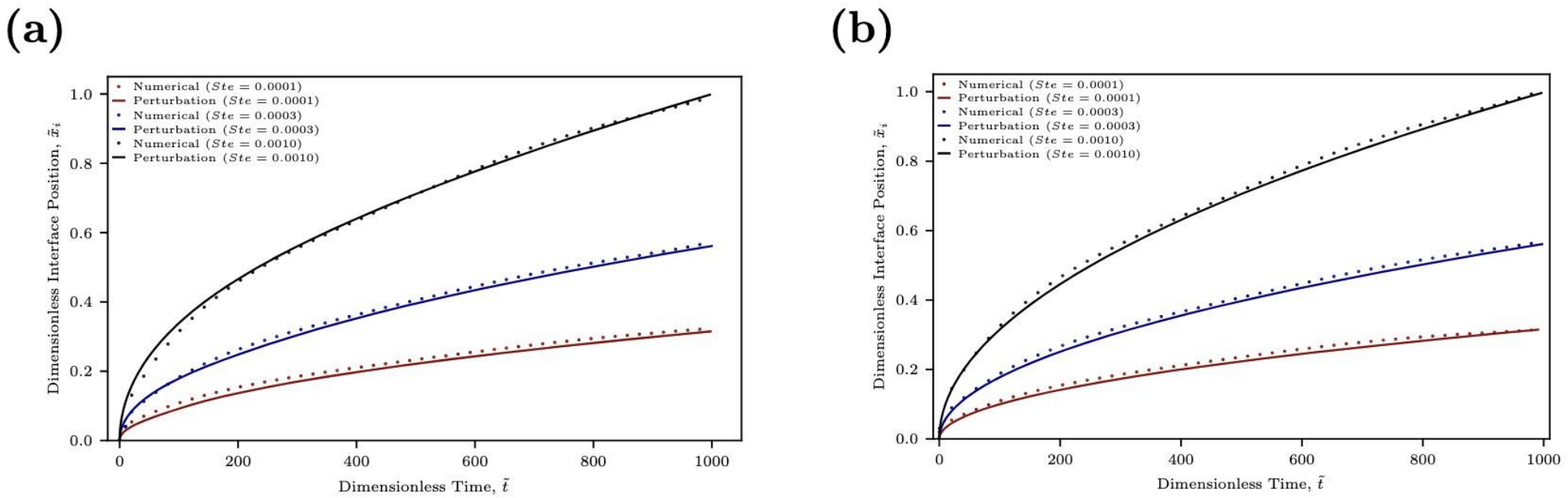


Figure 8: Dimensionless Interface Position vs. Dimensionless Time for Different Stefan Numbers with (a) Composite Solution with Diffusive Interface Condition ($\alpha = 1.0, \tau = 1.0$) and (b) Size-Dependent, Non-Diffusive Interface Condition ($\alpha = 1.0, \tau = 1.0$).

Conclusion and Future Work

What We Developed:

- Perturbation series expansion solutions to the hyperbolic Stefan problem to improve implementation facility.
- Clear outline for relevant numerical scheme (TVD MacCormack).

Future Work / Applications:

- Extension to different geometries, particularly spherical and cylindrical, relevant for nanoscale melting processes.
- Equations can be applied when observing phase change on the micro-scale in similar geometries.

Thank You!

Questions?

Example Applications:

In the context of surface ablation of semiconductors:

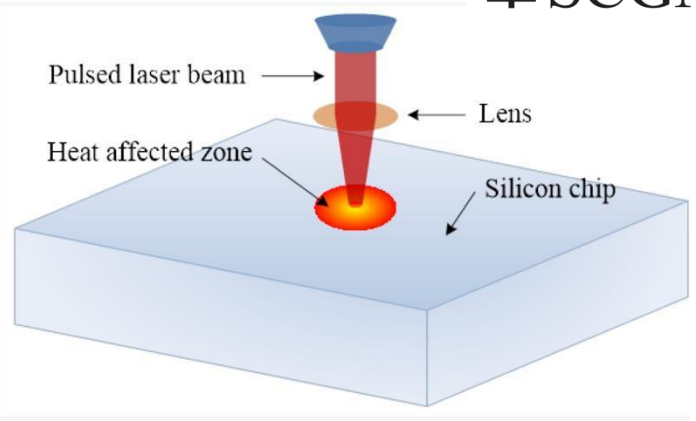


Figure 9: Can locally approximate the phase change induced by ultrafast laser heating to be a 1D Hyperbolic Stefan proble. (Kou et al., *Coatings*, 2023)

In the context of cell phase change:

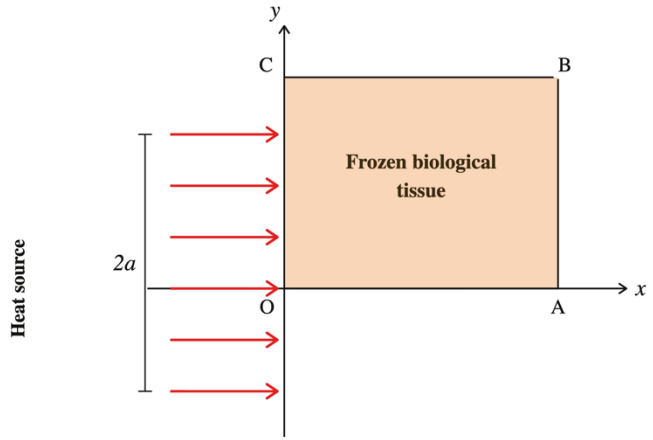


Figure 10: Can model smallscale biological phase changes, i.e. applications in Cryosurgery (Li et al., *Journal of Thermal Stresses*, 2023)